

INDIAN INSTITUTE OF TECHNOLOGY KHARAGPUR

1. Name of the Academic Unit: Department of Electrical Engineering
2. Subject Name: Optimization Models and Algorithms L-T-P: 3-0-0 Credits: 3
3. Pre-requisites: NIL
4. Syllabus and reference books:

Syllabus:

Introduction, Background, Linear and Quadratic Models, Sparsity and Optimization Models, Convex Models and Semidefinite Programming, Convex Models and Semidefinite Programming, Overview on some other optimization models, and Optimization Algorithms

Reference Books:

- 1) Practical Methods of Optimization, R. Fletcher, Wiley.
- 2) Optimization: Insights and Applications, J Brinkhuis, V. Tikhomirov, New Age International Publishers, 2010.
- 3) Introduction to Nonlinear Optimization, A Beck, SIAM, 2014.
- 4) Optimization Models, G. C. Calafiore and L. E. Ghaoui, Cambridge University Press, 2014.
- 5) Convex Optimization, S Boyd and L Vandenberghe, Cambridge University Press, 2004.

5. Lecture-wise break-up:

Sl. No.	Topic	No. of lectures
1.	Introduction Classes of optimization models, Unconstrained and constrained optimizations, Optimization in Electrical Engineering	1
2.	Backgrounds Local and global optimal solutions, Conditions for optimality, Vector spaces, Symmetric matrices, Singular Value Decomposition	4
3.	Linear and Quadratic Models Linear programming, Simplex method, Duality, l_1 and l_∞ norm regression problems, Least squares, Minimum-norm solutions, Applications	8
4.	Sparsity and Optimization Models Sparsity and Compression, l_1 norm and sparse solution, Sparse regression, LASSO regression, Applications	4

5.	Convex Models and Semidefinite Programming Convex sets and functions, Convex models, Duality, Convex relaxation	8
6.	Overview on some other optimization models Integer Programming, Network Programming	3
7.	Optimization Algorithms Descent algorithms, Stochastic Gradient Descent algorithm, Interior point algorithms, Proximal gradient algorithm.	12
Total number of hours		40